

LOW CODE

EXPERIMENT – 1 Comparative study of low-code development versus traditional application development approaches.

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AIM

To conduct a comparative study of low-code development and traditional application development approaches, focusing on their development processes, time, cost, complexity, flexibility, and ease of use.

INTRODUCTION TO TRADITIONAL APPROACH

Traditional application development is a conventional approach to building software applications by writing program code manually using programming languages such as Java, Python, C++, or C#. It generally follows a structured software development life cycle, beginning with requirement analysis and system design, followed by coding, testing, deployment, and maintenance.

In this approach, developers have complete control over the application's functionality, architecture, user interface, database, and integrations. However, it requires skilled programmers and usually takes more time and resources to develop and maintain applications. Traditional development is suitable for complex and highly customized applications where flexibility and detailed control are important.

INTRODUCTION TO LOW CODE DEVELOPMENT

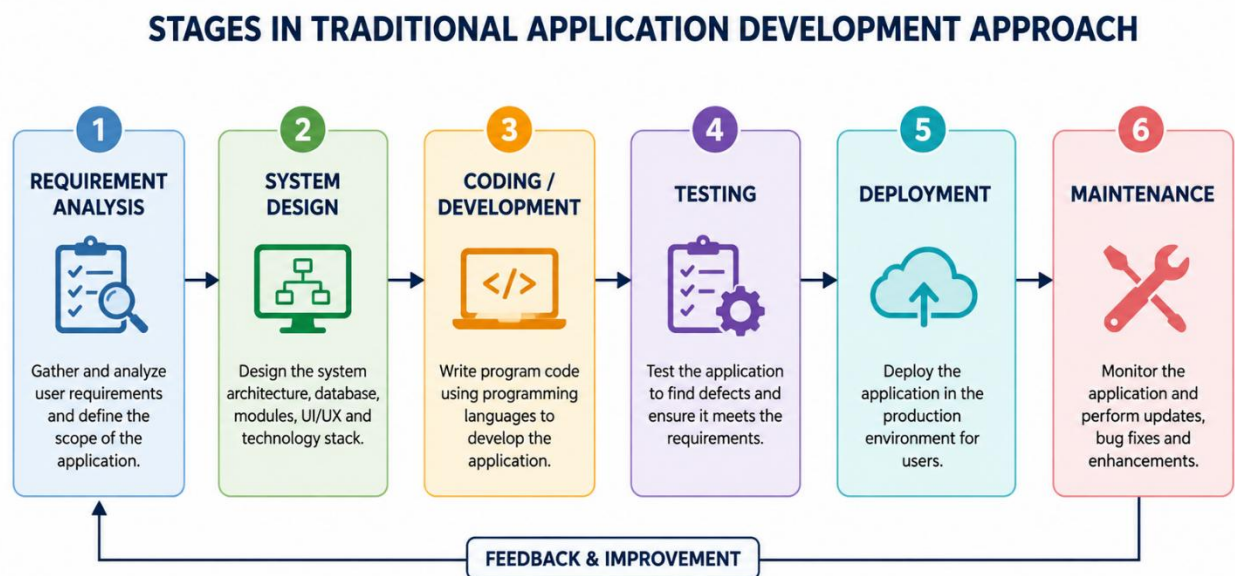
Low-code development is a software development approach that allows applications to be created with minimal manual coding. It uses visual development environments, drag-and-drop components, pre-built templates, and ready-made modules to simplify the application development process.

Low-code platforms enable developers to design user interfaces, workflows, databases, and integrations using visual tools instead of writing large amounts of code. This reduces development time and makes application

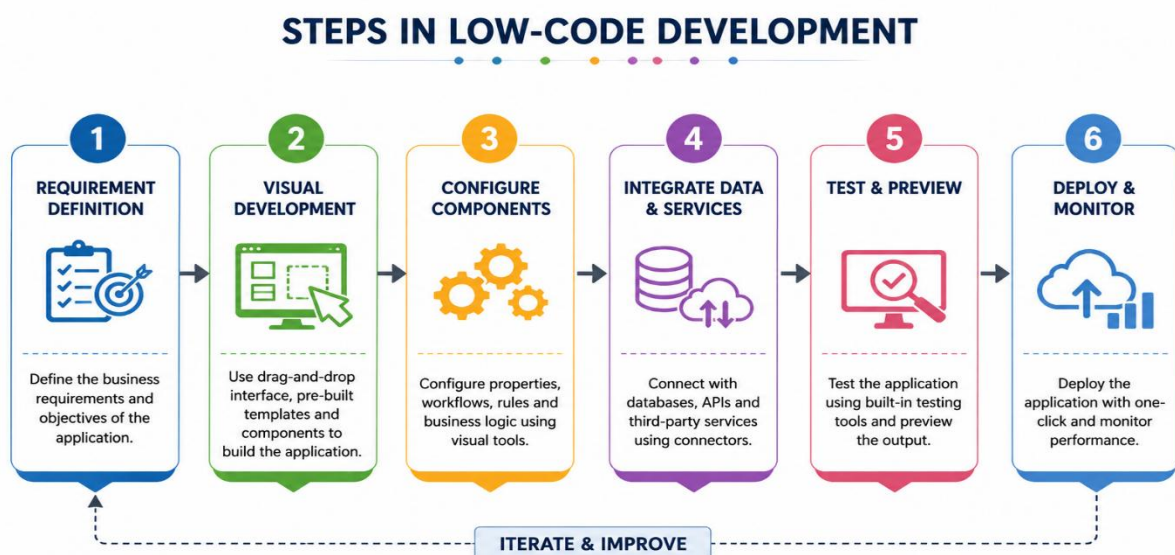
development easier for both professional developers and users with limited programming knowledge.

Low-code development is useful for rapidly building business applications, prototypes, automation solutions, and simple to moderately complex applications. However, it may provide less flexibility and customization compared to traditional application development.

STAGES IN TRADITIONAL APPROACH



STEPS IN LOW CODE



COMPARITIVE ANALYSIS FOR TRADITIONAL AND LOW CODE

Criteria	Traditional Development	Low code development
Development Method	Requires manual coding using programming languages.	Uses visual tools, drag-and - drop components, and minimal coding.
Development Time	Takes more time to design, code and test	Faster due to pre-built components and templates.
Programming skills	Requires strong programming and technical knowledge.	Requires less programming knowledge.
Customization	Provides high flexibility and complete customization.	Customization may be limited by the platform.
Cost and maintenance	Usually requires more resources and manual maintenance.	Reduces development effort and simplifies maintenance.

RESULT

The comparative study of traditional application development and low-code development was successfully completed. Traditional development provides greater flexibility and customization but requires more time and programming skills. Low-code development enables faster application development with less coding effort and simplified maintenance. Thus, the suitable approach can be selected based on the application's requirements, complexity, time, and available resources.